



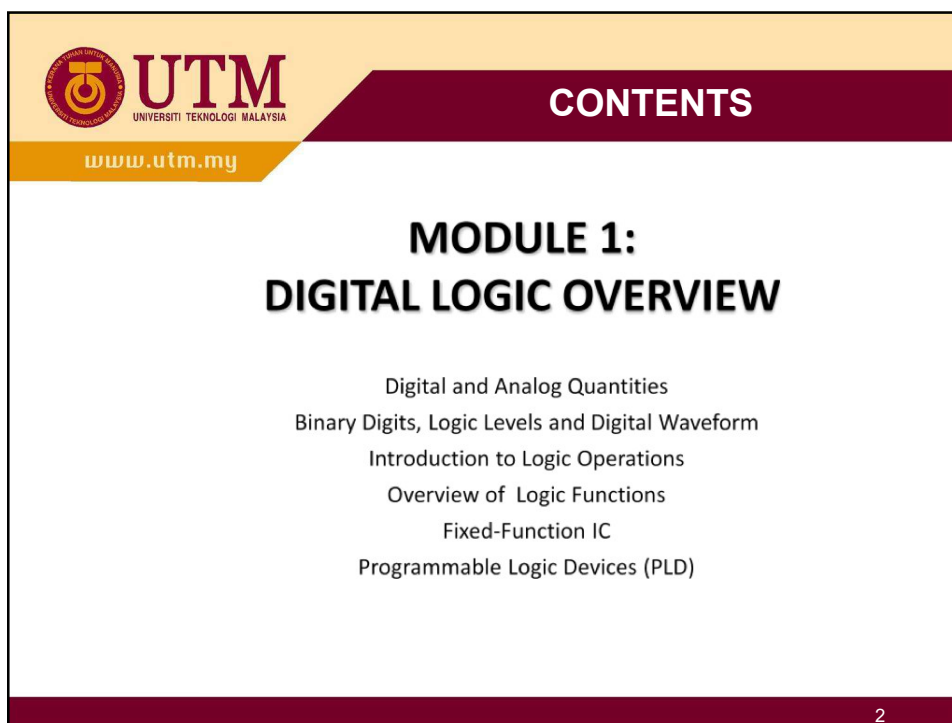
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SCSR1013 DIGITAL LOGIC

**MODULE 1:
INTRODUCTORY CONCEPTS**

2023/2024-1

FACULTY OF COMPUTING



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CONTENTS

**MODULE 1:
DIGITAL LOGIC OVERVIEW**

- Digital and Analog Quantities
- Binary Digits, Logic Levels and Digital Waveform
- Introduction to Logic Operations
- Overview of Logic Functions
- Fixed-Function IC
- Programmable Logic Devices (PLD)

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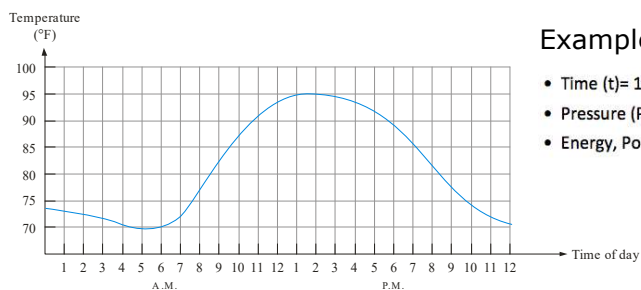
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Digital and Analog Quantities

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Analog quantities

- Most natural quantities that we see are **analog** and vary continuously.
- Analog systems can generally handle higher power than digital systems.



Example:

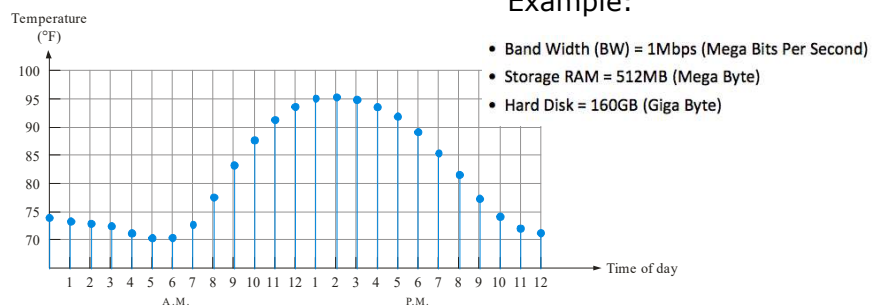
- Time (t) = 10.16s (second)
- Pressure (P) = 220.10KPa (Kilo Pascal)
- Energy, Power = 100.5KW (Kilo Watts)

Resource: Floyd, Digital Fundamentals, 10th Edition

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Digital quantities

Example:



- Digital systems can **process**, **store**, and **transmit** data more efficiently but can only assign discrete values (**discontinuous**) to each point.

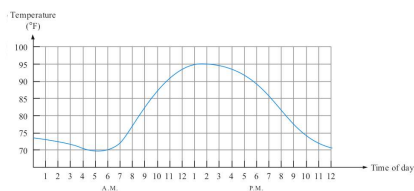
Resource: Floyd, Digital Fundamentals, 10th Edition

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Analog vs Digital

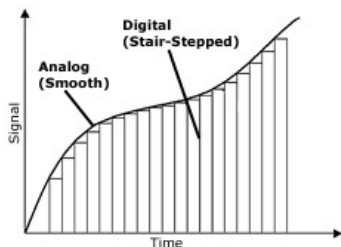
Analog

- Use base 10 (decimal)
- Represented by 10 different level: 0, 1, 2, 3, 4, 5, 6, 7, 8, and 9.
- Analog system: A combination of devices that manipulate values represented in analog form

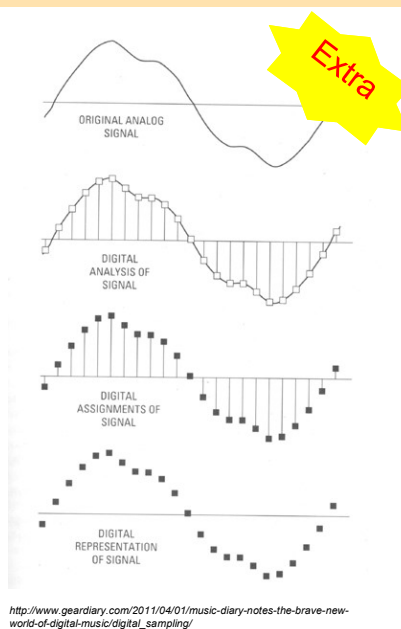


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Example of sampling analog-to-digital (frequency at least 2 times higher than analog)

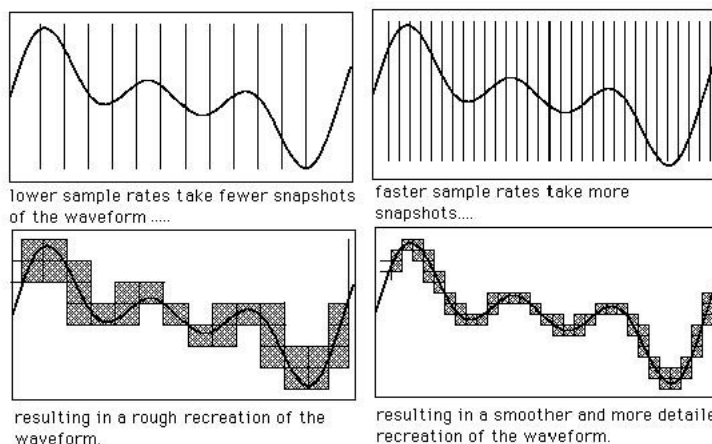


<http://www.blazeaudio.com/howto/bg-digital.html>



http://www.geardairy.com/2011/04/01/music-diary-notes-the-brave-new-world-of-digital-music/digital_sampling/

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<http://musicandcomputers306.blogspot.com/2010/10/waveforms-ad-conversion-sampling.html>

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Self-Test: Which of the following belong to analog system?

Extra

(a)  (b)  (c) 

(d)  (e)  (f) 

Resource: Google searched

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Answer

Extra

(a)  (c) 

(e) 

Resource: Google searched

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Extra

Example: Analog systems



Resource: Google searched

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Extra

Example: Digital systems



Resource: Google searched

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Digital

- Digital technology is relatively new compared to analog technology, but a lot of analog systems has been changed to a digital systems, Examples:
 - Computers
 - Manufacturing systems
 - Medical Science
 - Transportation
 - Entertainment
 - Telecommunications



*DSL-2320B (ADSL Modem)

Resource: <http://www.wirelessnetworkproducts.com/dsl-2320b.aspx>

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Digital

Exercise: Match the picture to which digital application system it belong to.



(b)



(e)



(a)



(d)



(f)



(c)

- (a) Computers
- (b) Manufacturing systems
- (c) Medical Science
- (d) Transportation
- (e) Entertainment
- (f) Telecommunications

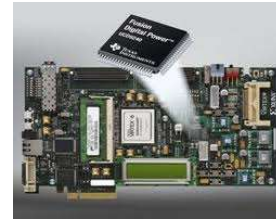
Resource: <http://www.wirelessnetworkproducts.com/dsl-2320b.aspx>

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The Digital Advantages

- Ease of design
- Ease of storage
- Accuracy and precision are easier to maintain
- Programmable operation
- Less affected by noise
- Ease of fabrication on IC chips

- Thus, the digital systems is more efficient and reliable for:
- Data Processing
 - Data Transmission
 - Data Storage

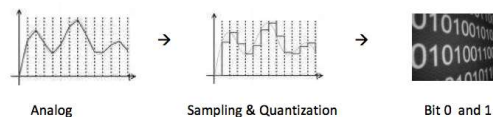


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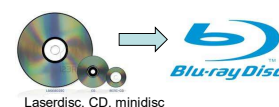
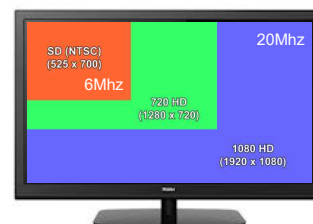
Digital Disadvantages

- Greater bandwidth
- Sampling error

Sampling Error (Quantization Error): Is derived from Analog to Digital Conversion Process:



- Compatibility with existing analog systems
- Short product half life



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Analog and Digital Systems

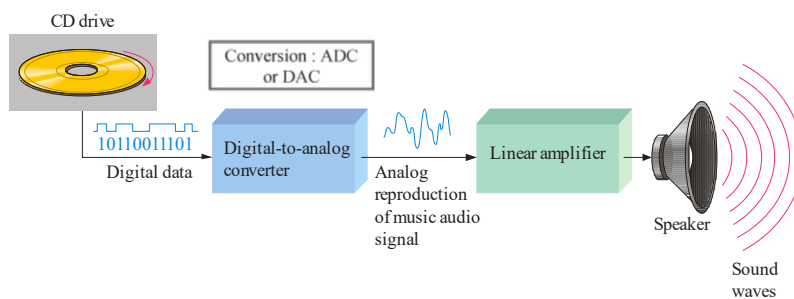
- Many systems use a **mix** of analog and digital electronics to take advantage of each technology.
- A typical CD player accepts digital data from the CD drive and converts it to an analog signal for amplification.



<http://www.8-echo.com/2009/11/14/bose-wave-music-system-and-multi-cd-changer-bundle.html>
<http://cdn-static.zdnet.com/l/story/61/18/006128/31929466-2-440-overview-1.gif>

Resource: Floyd, Digital Fundamentals, 10th Edition

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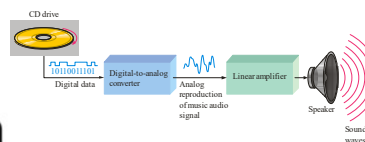


1. Convert digital sound (CD) to analog
2. Process (amplify) the analog information
3. Convert the analog signal to sound

Resource: Floyd, Digital Fundamentals, 10th Edition

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Hybrid System



- The audio CD is a typical hybrid (Analog & Digital) system.
 - Analog sound is converted into analog voltage using a microphone.
 - Analog voltage is changed into digital through an ADC in the recorder.
 - Digital information is stored on the CD .
 - At playback the digital information is changed into analog by a DAC in the CD player.
 - The analog voltage is amplified and used to drive a speaker that produces the original analog sound.

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Conversion:



<http://www.idt.com/products/data-converters>

Analog to Digital Converter (ADC):

- Convert analog signal into digital signal using process such as sampling, quantization process and digital conversion.
- Error will occur during the sampling and quantization, hence loss of information can happen.

Digital to Analog Converter (DAC):

- Needed if the speaker is using analog system.
- Need to convert the digital data to analog signal in order for the speaker works properly and the sound can heard by human.



<http://thesoundviewstudio.com/audio-rentals.html>

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Digits, Logic Levels and Digital Waveform

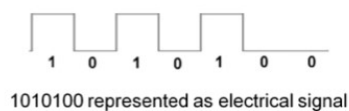
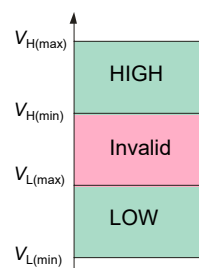
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Binary digits and logic levels

Digital electronics uses circuits that have two states, which are represented by two different voltage levels:

- HIGH (bit 1)
- LOW (bit 0)

A bit can have the value of either a 0 or a 1, depending on if the voltage is **HIGH** or **LOW**.



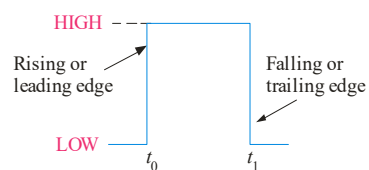
Resource: Floyd, Digital Fundamentals, 10th Edition

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- **Digital waveforms** change between the **LOW** and **HIGH** levels.
- A positive going pulse is one that goes from a normally **LOW** logic level to a **HIGH** level and then back again.
- **Digital waveforms** are made up of a series of pulses.

Resource: Floyd, Digital Fundamentals, 10th Edition

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(a) Positive-going pulse

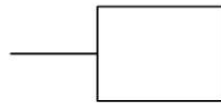
**Positive Logic
(active high)**

High = 1 (Bit 1)

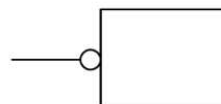
Low = 0 (Bit 0)

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Symbols to show the input state of "active high" and "active low":



"active high"



"active low"

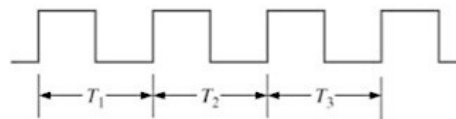
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Digital Waveforms

- Two type of squarewave

- Periodic

- The signal keep on repeating after a period of time



$$\text{Period} = T_1 = T_2 = T_3 = \dots = T_n$$

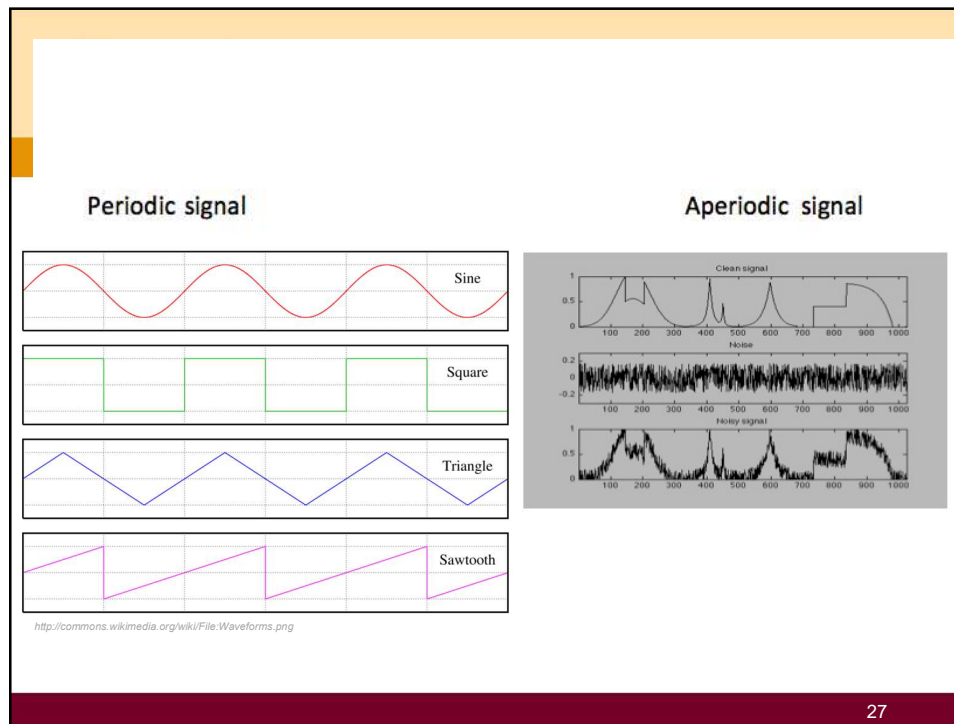
$$\text{Frequency} = \frac{1}{T}$$

- Non-Periodic / Aperiodic

- Doesn't have a period



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Periodic Signal Parameter

- Frequency (f) is the rate at which the signal repeat itself at a fixed interval. Is measured in cycles per second or Hertz (Hz)

$$f = \frac{1}{T} \text{ Hz}$$
- Period (T) is the time from the edge of one pulse to the corresponding edge of the next pulse. Is measured in second

$$T = \frac{1}{f} \text{ seconds}$$

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Example:

- clock frequency : $f = 100\text{Hz}$,
so, period : $T = 1/100\text{Hz} = 0.01\text{s} = 10 \times 10^{-3} = 10\text{ ms}$

$\text{s} \rightarrow \text{ms} (\times 10^3)$
 $\text{ms} \rightarrow \text{s} (\times 10^{-3})$

Some examples of periodic signal display on the oscilloscope:



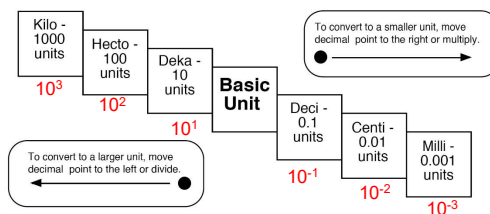
(a) Square waveform



(b) Sinusoid waveform

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Metric Conversion Chart



Unit Conversion

- | | |
|------------------------|-------------------------------|
| ♦ Kilo (K) = 10^3 | ♦ Mili (m) = 10^{-3} |
| ♦ Mega (M) = 10^6 | ♦ Micro (μ) = 10^{-6} |
| ♦ Giga (G) = 10^9 | ♦ Nano (n) = 10^{-9} |
| ♦ Tera (T) = 10^{12} | ♦ Piko (p) = 10^{-12} |

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- ♦ Mili (m) = 10^{-3}
- ♦ Micro (μ) = 10^{-6}
- ♦ Nano (n) = 10^{-9}
- ♦ Piko (p) = 10^{-12}

Example : $f = 100\text{KHz}$, So

$$\begin{aligned} T &= 1/f \\ &= 1/(100 \times 10^3 \text{Hz}) \\ &= 0.01 \times 10^{-3} \text{ s} \\ &= 0.01 \text{ms} \\ &= 10 \mu\text{s} \end{aligned}$$

$$\begin{aligned} &= (0.01 \times 10^{-3}) \text{ s} \times 10^3 \\ &= (0.01 \times 10^{-3+3}) \text{ ms} \\ &= (0.01 \times 10^0) \text{ ms} \\ &= 0.01 \text{ ms} \end{aligned}$$

$$\begin{aligned} &= (0.01 \times 10^{-3}) \text{ s} \times 10^6 \\ &= (0.01 \times 10^{-3+6}) \mu\text{s} \\ &= (0.01 \times 10^3) \mu\text{s} \\ &= 10 \mu\text{s} \end{aligned}$$

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Exercise 1.1 : Calculate the frequency of signals if time period are given as the following:

- $10\text{ms} = \underline{\hspace{2cm}} \text{Hz}$
- $100\mu\text{s} = \underline{\hspace{2cm}} \text{KHz}$
- $100\text{ns} = \underline{\hspace{2cm}} \text{MHz}$
- $1000\text{ps} = \underline{\hspace{2cm}} \text{GHz}$

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Solution 1.1 :

(Convert to second)

$$\text{a) } 10\text{ms} = \left(\frac{10}{10^3}\right)s = \left(\frac{1}{10^2}\right)s = 0.01s$$

$$\text{b) } 100\mu s = \left(\frac{100}{10^6}\right)s = \left(\frac{1}{10^4}\right)s = 0.0001s$$

$$\text{c) } 100\text{ns} = \left(\frac{100}{10^9}\right)s = \left(\frac{1}{10^7}\right)s = 10^{-7}s$$

$$\text{d) } 1000\text{ps} = \left(\frac{1000}{10^{12}}\right)s = \left(\frac{1}{10^9}\right)s = 10^{-9}s$$

$$f = \frac{1}{0.01} = 100\text{Hz}$$

$$f = \frac{1}{0.0001} = 1 \times 10^4 \text{Hz} = \frac{1 \times 10^4}{10^3} = 10\text{KHz}$$

$$f = \frac{1}{10^{-7}} = 1 \times 10^7 \text{Hz} = \frac{1 \times 10^7}{10^6} = 10\text{MHz}$$

$$f = \frac{1}{10^{-9}} = 1 \times 10^9 \text{Hz} = \frac{1 \times 10^9}{10^9} = 1\text{GHz}$$

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Exercise 1.2 : Calculate the time period of signals if the frequencies are given as the following:

$$\text{a) } 1000\text{KHz} = \underline{\hspace{2cm}} \mu s$$

$$\text{b) } 100\text{MHz} = \underline{\hspace{2cm}} ns$$

$$\text{c) } 1000\text{GHz} = \underline{\hspace{2cm}} ps$$

$$\text{d) } 100\text{THz} = \underline{\hspace{2cm}} ps$$

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Solution 1.2 :

(Convert to Hz)

$$\text{a) } 1000\text{KHz} = 1000 \times 10^3 \text{ Hz} = 10^6 \text{ Hz}$$

$$\text{b) } 100\text{MHz} = 100 \times 10^6 \text{ Hz} = 10^8 \text{ Hz}$$

$$\text{c) } 1000\text{GHz} = 1000 \times 10^9 \text{ Hz} = 10^{12} \text{ Hz}$$

$$\text{d) } 100\text{THz} = 100 \times 10^{12} \text{ Hz} = 10^{14} \text{ Hz}$$

$$T = \frac{1}{10^6} = 10^{-6} \text{ s} = 10^{-6} \times 10^6 = 1\mu\text{s}$$

$$T = \frac{1}{10^8} = 10^{-8} \text{ s} = 10^{-8} \times 10^9 = 10\text{ns}$$

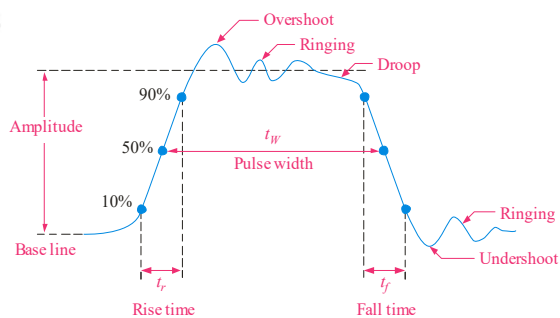
$$T = \frac{1}{10^{12}} = 10^{-12} \text{ s} = 10^{-12} \times 10^{12} = 1\text{ps}$$

$$T = \frac{1}{10^{14}} = 10^{-14} \text{ s} = 10^{-14} \times 10^{12} = 0.01\text{ps}$$

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Pulse Definition

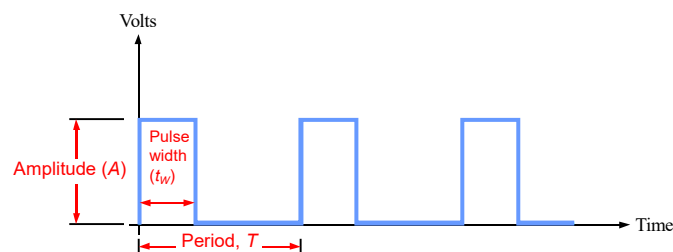
- Pulse is a rapid, transient change in the amplitude of a signal from a baseline value to a higher or lower value, followed by a rapid return to the baseline value.
- Pulse width (t_w): A measure of the duration of the pulse.
- Rise time and fall time is a measure of how fast the pulse change.

Resource: Floyd, Digital Fundamentals, 10th Edition

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Repetitive Pulse Waveform

- In addition to frequency and period, repetitive pulse waveforms are described by the **amplitude (A)**, **pulse width (t_w)** and **duty cycle**.
- Duty cycle is the ratio of t_w to T .



Resource: Floyd, Digital Fundamentals, 10th Edition

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Duty Cycle

- Duty cycle is the fraction of time that a system is in an "active" state (operated), defined as

$$\text{Duty cycle} = (t_w/T)100\%$$

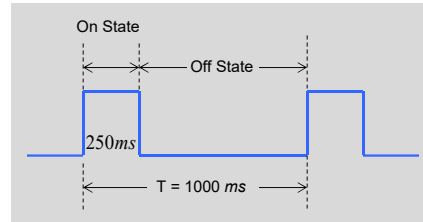


Example : a periodic digital waveform has a pulse width (t_w) 1ms and period time (T) 10ms, calculate duty cycle?

$$\text{Duty cycle} = 1\text{ms}/10\text{ms} * 100\% = 10\%$$

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Exercise 1.3: Given the duration or period of a system is 1000ms, determine the *on state* and *off state* of the system that operate with the ratio of duty cycle is 25%. Show your works.



Solution 1.3:

Duty cycle $\rightarrow$ on state

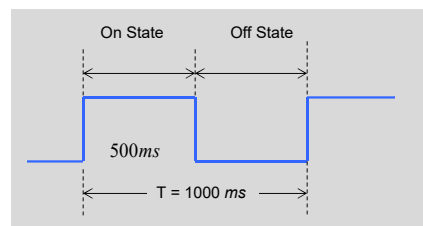
$$= 25\% \times 1000 \text{ ms} = \frac{25}{100} \times 1000 \text{ ms} = \frac{1}{4} \times 1000 \text{ ms} = 250 \text{ ms}$$

$$\text{Off state} : = 1000 \text{ ms} - 250 \text{ ms} = 750 \text{ ms}$$

(module: page 18)

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Exercise 1.4: Given the duration or period of a system is 1000ms, determine the *on state* and *off state* of the system that operate with the ratio of duty cycle is 50%. Show your works.



Solution 1.4:

Duty cycle $\rightarrow$ on state

$$= 50\% \times 1000 \text{ ms} = \frac{50}{100} \times 1000 \text{ ms} = \frac{1}{2} \times 1000 \text{ ms} = 500 \text{ ms}$$

$$\text{Off state} : = 1000 \text{ ms} - 500 \text{ ms} = 500 \text{ ms}$$

(module: page 18)

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Exercise 1.5: Given the *duty cycles* of a system is 40% for a duration of a system is 500ms.
 a) Calculate the pulse width of the system.
 b) Determine the *off state* of the system that operate with the ratio of duty cycle.
 Show your works.

Solution 1.5:

$$\text{a) } \text{DutyCycle} = \left(\frac{t_w}{T}\right)100$$

$$40 = \left(\frac{t_w}{500}\right)100$$

$$t_w = \frac{40(500)}{100} = 200s$$

$$\text{b) } \text{Period} = 500ms$$

$$\text{OnState} = t_w = 200ms$$

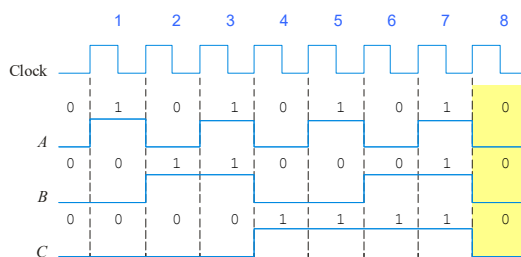
$$\therefore 500 - 200 = 300ms$$

(module: page 18)

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Timing diagram

A timing diagram is used to show the relationship between two or more digital waveforms,



At time 8, all A, B, and C **LOW**

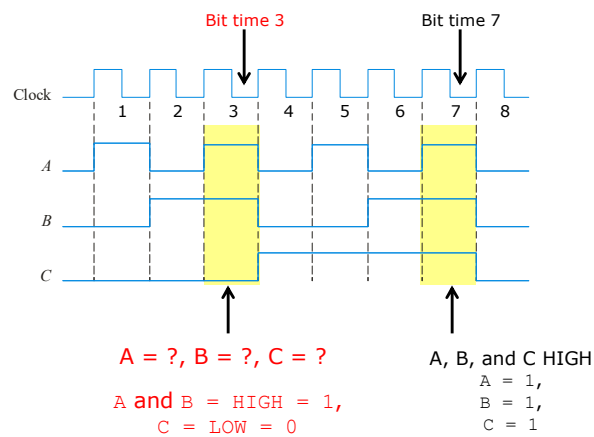
A diagram like this can be observed directly on a logic analyzer.



Resource: Floyd, Digital Fundamentals, 10th Edition

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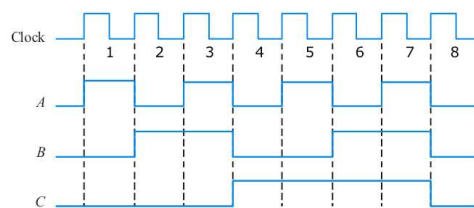
Example: Timing Diagram



Resource: Floyd, Digital Fundamentals, 10th Edition

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Example: Timing Diagram



Exercise: Complete the truth table.

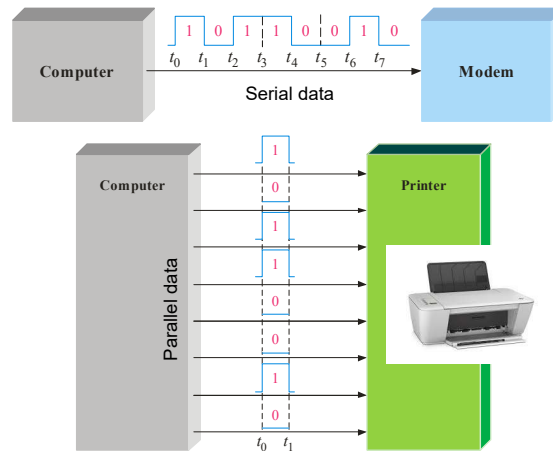
Clock (↑)	Input		Output
	A	B	C
1	1	0	0
2	0	1	0
3	1	1	0
4	0	0	1
5	1	0	1
6	0	1	1
7	1	1	1
8	0	0	0

Resource: Floyd, Digital Fundamentals, 10th Edition

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Data Transfer

Data can be transmitted by either **serial** transfer or **parallel** transfer.



Resource: Floyd, Digital Fundamentals, 10th Edition

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Introduction to Logic Operations

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Basic logic function

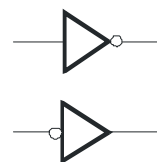
AND True only if **all** input conditions are true.



OR True only if **one or more** input conditions are true.



NOT Indicates the **opposite** condition (inverter).

Resource: Floyd, Digital Fundamentals, 10th Edition

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Logic Gates: NOT

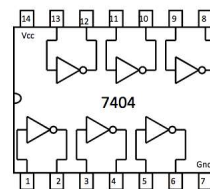


NOT operation

Truth table shows the relationship between output and the input.

Truth Table for NOT

X	Z
0	1
1	0



7404 IC six inverters

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Logic Gates: AND

HIGH (1) — HIGH (1)
HIGH (1) —

LOW (0) — LOW (0)
HIGH (1) —

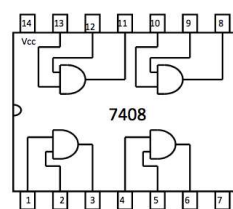
HIGH (1) — LOW (0)
LOW (0) —

LOW (0) — LOW (0)
LOW (0) —

AND operation

Truth Table AND

X	Y	Z
0	0	0
0	1	0
1	0	0
1	1	1



7408 IC four (Quad) AND gates

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Logic Gates: OR

HIGH (1) — HIGH (1)
HIGH (1) —

LOW (0) — HIGH (1)
HIGH (1) —

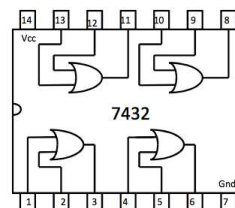
HIGH (1) — HIGH (1)
LOW (0) —

LOW (0) — LOW (0)
LOW (0) —

OR operation

Truth Table OR

X	Y	Z
0	0	0
0	1	1
1	0	1
1	1	1

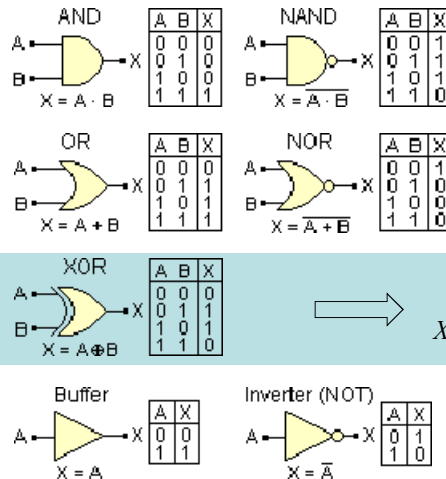


7432 integrated circuit provides four (Quad) two-inputs OR gates

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Logic Gates:

Summary



Exclusive-NOR gate

Input_A Input_B Output

A	B	Output
0	0	1
0	1	0
1	0	0
1	1	1

Equivalent gate circuit

Input_A Input_B Output

Resource: <http://www.chem.uoa.gr/applets/appletgates/Images/Image1.gif>

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Overview of Logic Functions

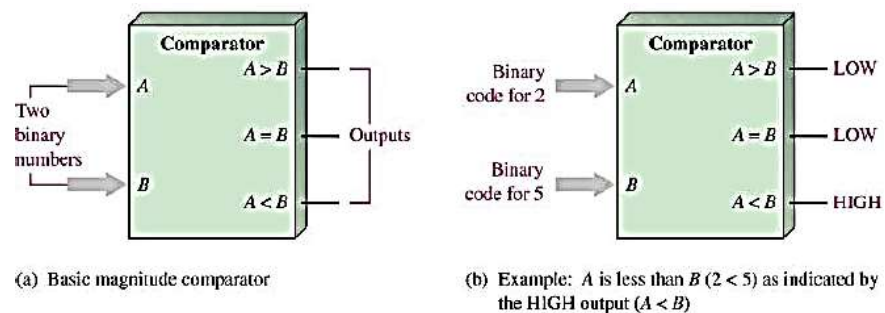
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Basic Logic Functions

- ❑ Any digital systems has one or more of the following function.
 - ❑ This functions are built from the basic gates.
 - Comparison Function
 - Arithmetic Functions
 - Code conversion function
 - Encoding function
 - Decoding function
 - Data selection function
 - Data storage function
 - Counting function

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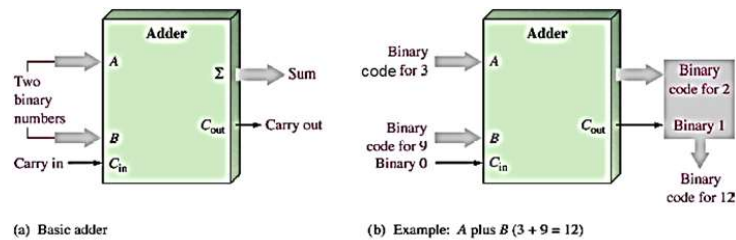
Comparison Function



54

Arithmetic Functions

- Adder



- Subtractor
- Multiplier
- Division

Resource: Floyd, Digital Fundamentals, 10th Edition

55

All the other arithmetic operations can be derived from adder:

- ☐ Subtraction is and addition of negative number such as $A - B = A + (-B)$
- ☐ Multiplication is a repeated addition such as $A * 3 = A + A + A$
- ☐ Division is a repeated subtraction which is a repeated addition such as $6/3 = 6 - 3 - 3 = 6 + (-3) + (-3)$
 - subtract until the remainder = 0
 - total number of subtraction = 2 which is the answer

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Code Conversion Function

- A code is a set of bits arranged in a unique pattern and used to represent specified information.
 - Examples : BCD, ASCII
- The usage of codes allow a faster and more efficient data processing.

57

http://www.ehow.com/how_7162480_
<http://facebooksmileysinfo.com/wp-content/uploads/2012/04/Smiley-Facebook-emoticons.jpg>
<http://depositphotos.com/2746252/stock-illustration-Arabic-alphabet.html>
<http://allenmathblog.files.wordpress.com/2012/01/integers.jpg>

58

[http://msdn.microsoft.com/en-us/library/60ec988f\(v=vs.80\).aspx](http://msdn.microsoft.com/en-us/library/60ec988f(v=vs.80).aspx)

Ctrl	Dec	Hex	Char	Code	Dec	Hex	Char	Dec	Hex	Char	Dec	Hex	Char
^@	0	00		NUL	32	20	!	64	40	@	96	60	z
^A	1	01		SOH	33	21	!"	65	41	A	97	61	[
^B	2	02		STX	34	22	!"#	66	42	B	98	62	\
^C	3	03		ETX	35	23	!"#\$	67	43	C	99	63	]
^D	4	04		END	36	24	!"#\$%	68	44	D	100	64	^
										E	101	65	_
										F	102	66	`
										G	103	67	a
										H	104	68	b
										I	105	69	c
										J	106	6A	d
										K	107	6B	e
										L	108	6C	f
										M	109	6D	g
										N	110	6E	h
										O	111	6F	i
										P	112	70	j
										Q	113	71	k
										R	114	72	l
										S	115	73	m
										T	116	74	n
										U	117	75	o
										V	118	76	p
										W	119	77	q
										X	120	78	r
										Y	121	79	s
										Z	122	7A	t
										[	123	7B	u
										\	124	7C	v
										]	125	7D	w
										^	126	7E	x
										_	127	7F	y

To send this:

- 😊 Smile
- 😮 Surprised
- 😉 Wink
- 😞 Confused
- 😭 Crying
- 🔥 Hot
- 👼 Angel
- 🙊 Don't tell anyone
- 🤓 Nerd
- 🗨️ Secret telling
- 😕 I don't know
- 🎉 Party

Type this:

- :~) or :) Open-mouthed
- :-O or :o Tongue out
- ;-) or ;) Sad
- :-S or :s Disappointed
- :'(Embarrassed
- (H) or (h) Angry
- (A) or (a) Devil
- :-# Baring teeth
- 8-| Sarcastic
- :-* Sick
- :^ Thinking
- <:o Eye-rolling

Extra

ASCII Table:

Dec	Hex	Char
62	3E	>
63	3F	?
94	5E	^
95	5F	_

* ASCII code 127 has the code DEL. Under MS-DOS, this code has the same effect as ASCII 8 (BS). The DEL code can be generated by the CTRL + BKSP key.

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Encoding & Decoding Function

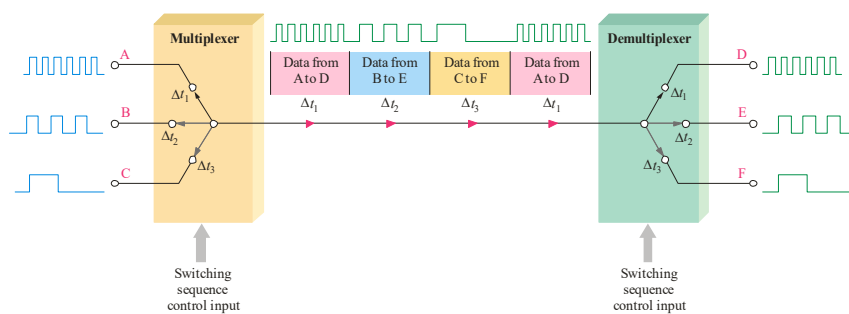
The encoding function

The decoding function

Resource: Floyd, Digital Fundamentals, 10th Edition

60

Data Selection Function: MUX & DeMUX



Resource: Floyd, Digital Fundamentals, 10th Edition

61

Problem:

Many inputs (e.g. A, B and C) wanted to use a single transmission line for their data transmission. How to make sure the data is transferred in a proper manner (issue of cost, synchronization, conflict, crash, loss?)

Source (A, B, C) and Destination (D, E, F)

$A \rightarrow D, B \rightarrow E, C \rightarrow F$

Solution:

MUX: select and permit only one device can use the line and transfer its data at one time.

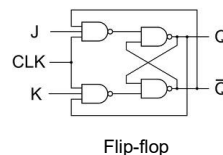
Data in the transmission line would be arranged as A, B, C

DEMUX: select and route the data to their originate destination

$A \rightarrow D, B \rightarrow E, C \rightarrow F$

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Data Storage Function



- Flip-flop stores a 1 or 0 only
- Registers
 - Formed by combining several flip-flops
 - 8-bit register → from 8 flip-flops
- Semiconductor Memories
 - e.g. RAM, ROM, Flash
- Magnetic/Optical Memories
 - For mass storage → e.g. hard disk, tape, DVD, Blu-Ray



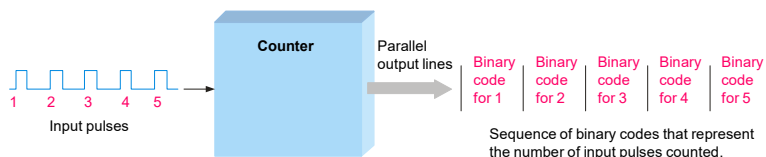
63

Counting Function

Examples:

- Counter
 - To count the occurrence at the input.
 - to initiate a controller after a certain count (period).

- Traffic light
- Washing machine
- Vending machine
- Xerox machine
- ATM machine
- etc.



Resource: Floyd, Digital Fundamentals, 10th Edition

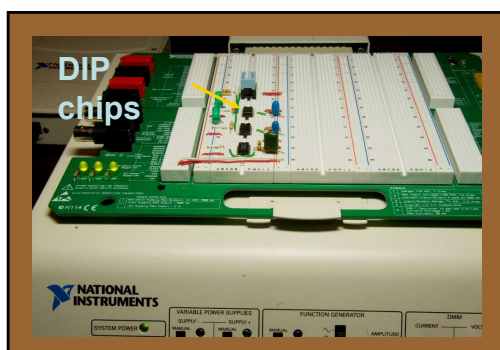
64

Fixed-Function Integrated Circuit (IC)

65

An example of laboratory prototyping is shown. The circuit is wired using DIP chips and tested.

In this case, testing can be done by a computer connected to the system.



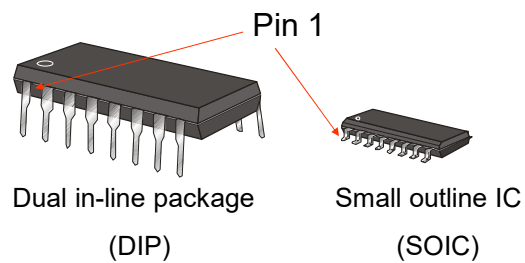
(Dual In-line Package)

Resource: Floyd, Digital Fundamentals, 10th Edition

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IC Packages

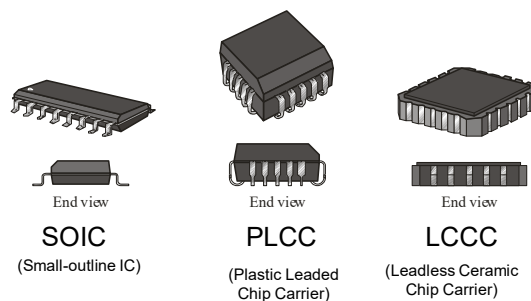
DIP chips and surface mount chips



Resource: Floyd, Digital Fundamentals, 10th Edition

67

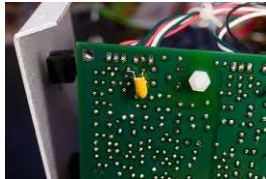
Other surface mount technology (SMT) packages:



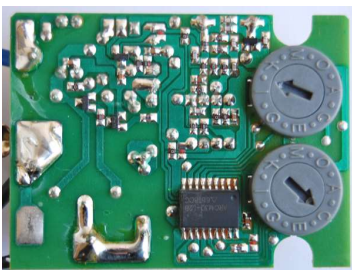
Resource: Floyd, Digital Fundamentals, 10th Edition

68

IC and conventional through-hole technology



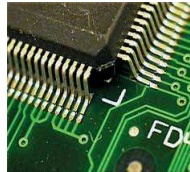
<http://aa7ee.wordpress.com/page/4/>



<http://jeelabs.org/tag/teardown/>

Printed Circuit Board (PCB)

Surface Mount PCB



<https://neuromorphs.net/ws2007/wiki/smd>

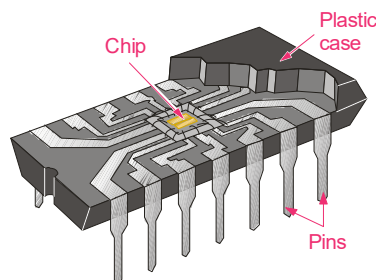


<http://www.pcb-manufacturers.co.uk/pcb-production-examples-c.html>

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Integrated circuit

Cutaway view of DIP (Dual-In-line Pins) chip:



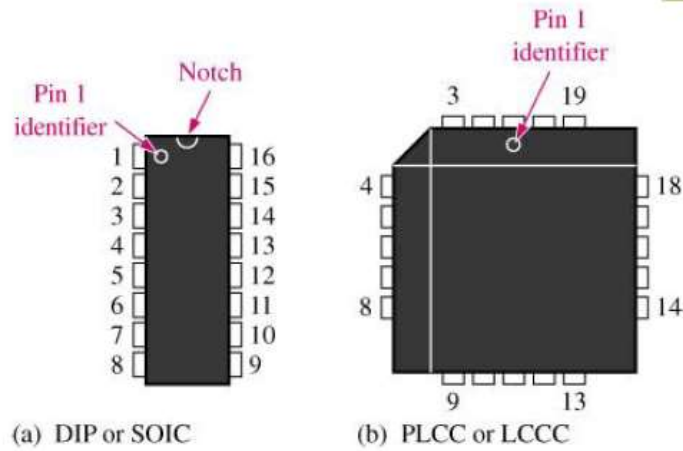
IC Packaging: Why we need packaging?

- To protect the IC (circuit)
- Have a pin system so that can connect to other circuit

Resource: Floyd, Digital Fundamentals, 10th Edition

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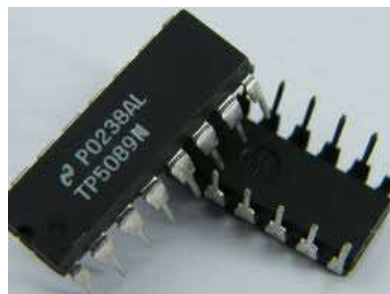
Pin Numbering



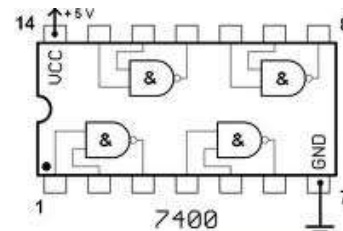
71



<http://www.rkonlinestore.co.uk/555-dual-timer-ic-16-pin-dip-pack-of-4-391-p.asp>



<http://www.ebay.com/itm/10pcs-IC-TP5089N-DIP-16-PIN-TP5089-310306081949>



<http://electroschematics.com/6529/7400-datasheet/>

Extra

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Complexity Classifications for Fixed-Function ICs

Module 1

- Small-scale integration (SSI)
have up to 12 gates on a single chip
- Medium-scale integration (MSI)
have from 12-99 gates on a single chip
- Large-scale integration (LSI)
have from 100-9999 gates on a single chip
- Very large-scale integration (VLSI)
have from 10,000-99,999 gates on a single chip
- Ultra large-scale integration (ULSI)
have from 100,000 and greater equivalent gates on a single chip



http://www.visual6502.org/images/263P/SL_263P_8404_chip1_package_top.jpg



<http://www.nysemagazine.com/lisicorp>

73

Integrated Circuit Technologies

Some examples of IC technologies:

- TTL (*Transistor-transistor Logic*)
- ECL (*Emitter-Coupled Logic*)
- CMOS (*Complementary Metal–Oxide–Semiconductor*)
- NMOS (*N-Type Metal–Oxide–Semiconductor*)
- BiCMOS (*Bipolar and Metal–Oxide–Semiconductor*)



CMOS –

<http://www.creativeplanetnetwork.com/dcp/news/cmos-technology-primer/40995>

74

Programmable Logic Devices (PLD)

75

Overview of PLD

□ Fixed function

- A specific logic function is contained in the IC (hardwired) and can never be changed.

□ PLD

- Logic function programmed by the user.
 - Some, can be reprogrammed many times.
- Advantage
 - More logic circuit can be 'stuffed' into much smaller area.
 - Certain PLD, design can be changed without rewiring or replacing components.
 - Can be implemented faster once the required programming language is mastered.

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Types of PLD

3 major types (SPLD, CPLD, FPGA)

1. Simple Programmable Logic Devices (SPLD)

- Can replace several fixed-function SSI or MSI
- First type available
- A few categories
 - PAL (programmable Array Logic)
 - GAL (Generic Array Logic)
 - PLA (Programmable Logic Array)
 - PROM (Programmable Read-Only memory)

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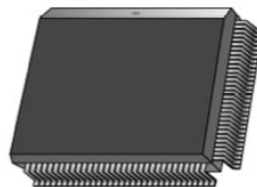
Types of PLD: CPLD

2. Complex Programmable Logic Devices (CPLD)

- Much higher capacity than SPLD (2-64 SPLD)
 - More complex logic circuits can be programmed
 - Typically in 44 – 160 pin package



(a) 84-pin PLCC package



(b) 128-pin PQFP package



Resource: http://upload.wikimedia.org/wikipedia/commons/thumb/a/a3/Altera_MAX_7128_2500_gate_CPLD.jpg/300px-Altera_MAX_7128_2500_gate_CPLD.jpg

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Types of PLD: FPGA

- 3. Field-Programmable Gate Arrays (FPGA)
 - Different internal organization than SPLD and CPLD
 - Greatest logic capacity
 - Consist of 64- thousands logic block (logic gate groups)
 - Classes
 - Fine grain (smaller logic block)
 - Coarse grain (large logic block)



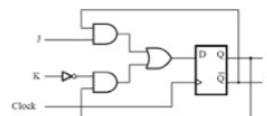
Resource: http://upload.wikimedia.org/wikipedia/commons/thumb/f/fa/Altera_StratixIVGX_FPGA.jpg/300px-Altera_StratixIVGX_FPGA.jpg

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PLD Programming

- Logic circuit entered using 2 basic method

- Graphical entry
 - schematic diagram



- Text-based entry (language based entry)

- Using Hardware Description Language (HDL)
 - Eg . ABEL, CUPL, WinCUPL
 - Becoming widely used especially for CPLD and FPGA
 - VHDL
 - Verilog

```

MODULE decoder
TITLE 'decoder'
A,B,C,D      pin      1,2,3,4;
W,X,Y,Z      pin      14,15,16,17;

equations
W=!B & C # !B & D # C & D # A;
X=!A & D # B # C;
Y=!A & !B & D # C;
Z=!B & C # D;

```

Resource: Floyd, Digital Fundamentals, 10th Edition

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